

All solutions must clearly show the steps or reasoning you used to arrive at your result. You will lose points for poorly written solutions or incorrect reasoning/explanations; answers given without explanation will not be graded.

Name:
Section:

1. **Fun with Taylor series.** Compute the following Taylor series about $x = 0$ to the given order, using the suggested methods:

- (a) (10 points) $f(x) = \frac{\sin x}{x}$ including terms up to x^2 , first from the *definition* of the Taylor series for $f(x)$, then by dividing the Taylor series for $\sin x$ by x .

From the definition of the Taylor series, $f(x) \approx f(0) + f'(0)x + \frac{f''(0)}{2}x^2 + \dots$. Since $\sin x/x = 0/0$ at $x = 0$, we need to use L'Hôpital's Rule, $\lim_{x \rightarrow 0} \frac{p(x)}{q(x)} = \lim_{x \rightarrow 0} \frac{p'(x)}{q'(x)}$, to compute the limits of $f(0)$, $f'(0)$, and $f''(0)$. First,

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1.$$

For the derivative,

$$\lim_{x \rightarrow 0} f'(x) = \lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{x^2} = \lim_{x \rightarrow 0} \frac{\cos x - x \sin x - \cos x}{2x} = \lim_{x \rightarrow 0} \left(-\frac{1}{2} \sin x \right) = 0.$$

Finally, for the second derivative,

$$\begin{aligned} \lim_{x \rightarrow 0} f''(x) &= \lim_{x \rightarrow 0} \frac{-x^2 \sin x - 2x \cos x + 2 \sin x}{x^3} \\ &= \lim_{x \rightarrow 0} \frac{-2x \sin x - x^2 \cos x - 2 \cos x + 2x \sin x + 2 \cos x}{3x^2} \\ &= \lim_{x \rightarrow 0} \left(-\frac{1}{3} \cos x \right) = -\frac{1}{3}. \end{aligned}$$

This yields

$$f(x) \approx 1 - \frac{x^2}{6} + \mathcal{O}(x^3).$$

The alternative (and much easier!) way is done by taking

$$f(x) = \frac{1}{x} \left(x - \frac{x^3}{6} + \mathcal{O}(x^5) \right) = 1 - \frac{x^2}{6} + \mathcal{O}(x^4),$$

which yields the same answer.

- (b) (10 points) $f(x) = \arctan(x)$ including terms up to x^3 , by using the fact that $\arctan(x) = \int \frac{dx}{1+x^2}$ and integrating a Taylor series.

$$\begin{aligned}\int_0^x \frac{dx'}{1+x'^2} &= \int_0^x dx' (1 - x'^2 + \mathcal{O}(x'^4)) \\ &= x - \frac{x^3}{3} + \mathcal{O}(x'^5).\end{aligned}$$

Or, for the indefinite integral one obtains $C + x - \frac{x^3}{3} + \mathcal{O}(x'^5)$.

In this case, we can find the constant by evaluating $\arctan(0) = 0$, so $C = 0$ and $\arctan(x) = x - \frac{x^3}{3} + \mathcal{O}(x^5)$.

- (c) (10 points) $f(x) = \exp(\sin(x))$ including terms up to x^4 , by plugging the series for $\sin x$ into the series for e^x .

$$\begin{aligned}f(x) &= e^{\sin x} \\ &= 1 + \sin x + \frac{\sin^2 x}{2!} + \frac{\sin^3 x}{3!} + \frac{\sin^4 x}{4!} + \dots \\ &= 1 + \left(x - \frac{x^3}{6} + \mathcal{O}(x^5)\right) + \frac{\left(x - \frac{x^3}{6} + \mathcal{O}(x^5)\right)^2}{2} + \frac{\left(x - \mathcal{O}(x^3)\right)^3}{6} + \frac{\left(x - \mathcal{O}(x^3)\right)^4}{24} + \dots\end{aligned}$$

Note that at this point, we only have to keep the leading terms in $\sin^3 x$ and $\sin^4 x$ because the next terms are $\mathcal{O}(x^5)$ or higher.

$$\begin{aligned}f(x) &= 1 + \left(x - \frac{x^3}{6}\right) + \left(\frac{x^2}{2} - \frac{x^4}{6}\right) + \frac{x^3}{6} + \frac{x^4}{24} + \mathcal{O}(x^5) \\ &= 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + \mathcal{O}(x^5).\end{aligned}$$

2. **Integrals with Taylor series.** (20 points) A common integral encountered in classical mechanics is

$$K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}},$$

which is known as an elliptic integral. If $k = 0$, this integral is simply $\int_0^{\pi/2} d\theta = \pi/2$. Compute the leading correction to this value for small k by expanding the integrand as a Taylor series. *Hint: your approximate integrand shouldn't have anything in the denominator.*

Let's write the denominator as $\sqrt{1-x}$ where $x = k^2 \sin^2 \theta$ and is small compared to 1. The Taylor series of this is given by $\frac{1}{\sqrt{1-x}} \approx 1 + x/2 + \dots$ which means that we can approximate

the total elliptic integral as

$$\begin{aligned}
 K(k) &= \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}} \\
 &\approx \int_0^{\pi/2} d\theta \left(1 + \frac{1}{2} k^2 \sin^2 \theta \right) \\
 &= \int_0^{\pi/2} d\theta \left(1 + \frac{1}{4} k^2 (1 - \cos 2\theta) \right) \\
 &= \frac{\pi}{2} \left(1 + \frac{k^2}{4} \right) - \frac{k^2}{8} \sin 2\theta \Big|_0^{\pi/2} \\
 &= \frac{\pi}{2} \left(1 + \frac{k^2}{4} \right).
 \end{aligned}$$

3. **Matrix multiplication and affine transformations.** In week 3 you saw that the matrix

$$M_A = \begin{pmatrix} \cos \theta & -\sin \theta & x^0 \\ \sin \theta & \cos \theta & y^0 \\ 0 & 0 & 1 \end{pmatrix}$$

transformed the first two components of a vector by rotating it through an angle θ and adding the vector $\vec{a} = (x^0, y^0)$. Another way to represent this transformation is an ordered pair $A = (R(\theta), \vec{a})$, where $R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ is the 2×2 matrix in the upper-left corner of the 3×3 matrix M_A . This is an *affine transformation* which acts on a 2-component vector $\vec{v}$ and returns $\vec{w} = R(\theta)\vec{v} + \vec{a}$.

- (a) (15 points) If we apply a second affine transformation $B = (R(\phi), \vec{b})$ to $\vec{w}$, what is the resulting 2-component vector $\vec{u}$ in terms of $R(\theta)$, $R(\phi)$, $\vec{a}$, $\vec{b}$, and $\vec{v}$?

Acting with the second affine transformation, B , on $\vec{w}$ yields $\vec{u}$. This is given by

$$\vec{u} = B\vec{w} = B[R(\theta)\vec{v} + \vec{a}] = R(\phi)R(\theta)\vec{v} + R(\phi)\vec{a} + \vec{b}.$$

- (b) (15 points) If we apply the transformations of part (a) in the opposite order (B applied to $\vec{v}$, then A applied to the result), what is the resulting two-component vector? You should find that it is *not* the same as the answer from part (a). Explain this difference geometrically (using a sketch if you find it helpful).

Applied in the other order, i.e. B first to $\vec{v}$ then A applied to the result, we

$$\vec{u} = A\vec{w} = A[R(\phi)\vec{v} + \vec{b}] = R(\theta)R(\phi)\vec{v} + R(\theta)\vec{b} + \vec{a}.$$

In both cases, we have two rotation matrices applied to $\vec{v}$, but (as shown in Discussion 6) rotation matrices commute, so $R(\theta)R(\phi)\vec{v} = R(\phi)R(\theta)\vec{v}$. The difference is in the vector that gets added: in part (a) it's $R(\phi)\vec{a} + \vec{b}$, while in part (b) it's $R(\theta)\vec{b} + \vec{a}$. In other words, the vector which gets added to the doubly-rotated $\vec{v}$ is either rotated- $\vec{a}$ plus $\vec{b}$, or rotated- $\vec{b}$ plus $\vec{a}$.

- (c) (20 points) Letting $\vec{b} = (x^1, y^1)$, write B as a 3×3 matrix M_B in the same form as M_A . Compute the matrix products $M_B M_A$ and $M_A M_B$, and explain how applying these matrices to $\vec{V} = (v^1, v^2, 1)$ agrees with your computations from (a) and (b).

$$M_B = \begin{pmatrix} \cos \phi & -\sin \phi & x^1 \\ \sin \phi & \cos \phi & y^1 \\ 0 & 0 & 1 \end{pmatrix}, \text{ so}$$

$$\begin{aligned} M_A M_B &= \begin{pmatrix} \cos \theta & -\sin \theta & x^0 \\ \sin \theta & \cos \theta & y^0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \phi & -\sin \phi & x^1 \\ \sin \phi & \cos \phi & y^1 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \cos(\theta)\cos(\phi) - \sin(\theta)\sin(\phi) & \sin(\theta)(-\cos(\phi)) - \cos(\theta)\sin(\phi) & x^1 \cos(\theta) + x^0 - y^1 \sin(\theta) \\ \sin(\theta)\cos(\phi) + \cos(\theta)\sin(\phi) & \cos(\theta)\cos(\phi) - \sin(\theta)\sin(\phi) & x^1 \sin(\theta) + y^1 \cos(\theta) + y^0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \cos(\theta + \phi) & -\sin(\theta + \phi) & x^1 \cos \theta - y^1 \sin \theta + x^0 \\ \sin(\theta + \phi) & \cos(\theta + \phi) & x^1 \sin \theta + y^1 \cos \theta + y^0 \\ 0 & 0 & 1 \end{pmatrix} \end{aligned}$$

Similarly,

$$\begin{aligned} M_B M_A &= \begin{pmatrix} \cos \phi & -\sin \phi & x^1 \\ \sin \phi & \cos \phi & y^1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta & x^0 \\ \sin \theta & \cos \theta & y^0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \cos(\theta)\cos(\phi) - \sin(\theta)\sin(\phi) & \sin(\theta)(-\cos(\phi)) - \cos(\theta)\sin(\phi) & x^0 \cos(\phi) + x^1 - y^0 \sin(\phi) \\ \sin(\theta)\cos(\phi) + \cos(\theta)\sin(\phi) & \cos(\theta)\cos(\phi) - \sin(\theta)\sin(\phi) & x^0 \sin(\phi) + y^0 \cos(\phi) + y^1 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \cos(\theta + \phi) & -\sin(\theta + \phi) & x^0 \cos \phi - y^0 \sin \phi + x^1 \\ \sin(\theta + \phi) & \cos(\theta + \phi) & x^0 \sin \phi + y^0 \cos \phi + y^1 \\ 0 & 0 & 1 \end{pmatrix} \end{aligned}$$

We can see that the rotational components of the two matrices (the upper left 2×2 block are identical as we expected. The differences are all in the third column where we see that acting first with M_B rotates $\vec{b} = (x^1, y^1)$ then adds $\vec{a}$, as opposed to rotating $\vec{a} = (x^0, y^0)$ and adding $\vec{b}$ when the matrices are multiplied in the opposite order.